Day 13: Random Variables and Probability Distributions
Turning Uncertainty into Numbers

100 Days of Data Science

Why Random Variables Are Important

In probability, outcomes are often unpredictable. Random variables convert uncertain outcomes into numerical values.

Machine Learning models work with numbers, not events.

Random variables bridge:

- Real-world uncertainty
- Mathematical modeling

What is a Random Variable?

A random variable is a function that assigns a numerical value to each outcome of a random experiment.

Example:

- Toss a coin
- Let X = number of heads

Possible values of X:

$$0 \text{ or } 1$$

Real-Life Example

Number of customers visiting a shop in one hour is a random variable.

It changes every hour but can be modeled using probability.

Types of Random Variables

Discrete Random Variable

Takes countable values.

Examples:

- Number of emails received
- Number of defective items
- Dice outcomes

Continuous Random Variable

Takes infinite possible values within a range.

Examples:

- Height of people
- Time taken to complete a task
- Temperature

Probability Distribution

A probability distribution describes how probabilities are assigned to values of a random variable.

It answers:

- Which values are likely?
- Which values are rare?

Discrete Distribution Example

Dice roll:

Value	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Continuous Distribution Example

Heights of people follow normal distribution.

Most values near average, fewer at extremes.

Why Distributions Matter in Data Science

- Model real-world data behavior
- Detect anomalies
- Choose appropriate ML algorithms

Random Variables in Machine Learning

ML models predict:

- Output values as random variables
- Probabilities over outcomes

Examples:

- Classification probabilities
- Regression prediction ranges

Common Mistakes

- Treating continuous data as discrete
- Ignoring distribution shape
- Assuming uniform distribution always

Final Takeaway

Random variables transform uncertainty into measurable quantities.

Probability distributions describe how real-world data behaves.

These concepts are the backbone of Machine Learning models.